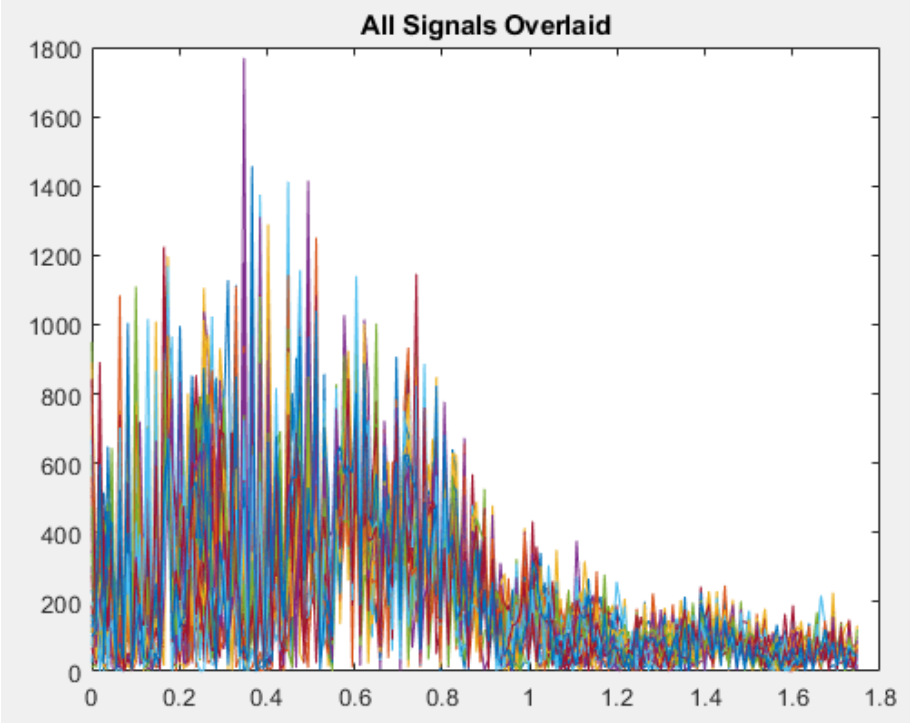
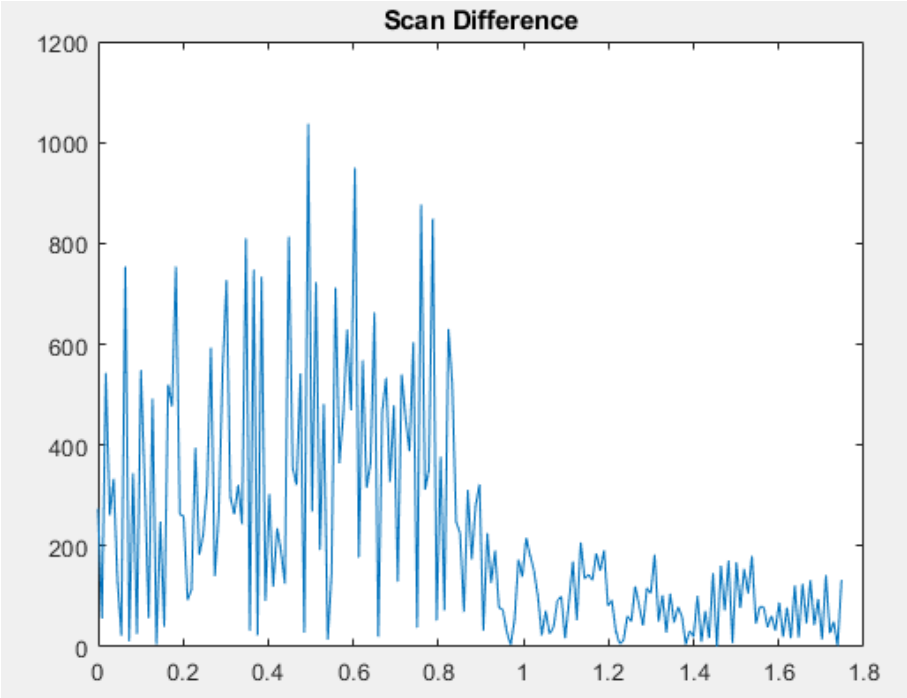
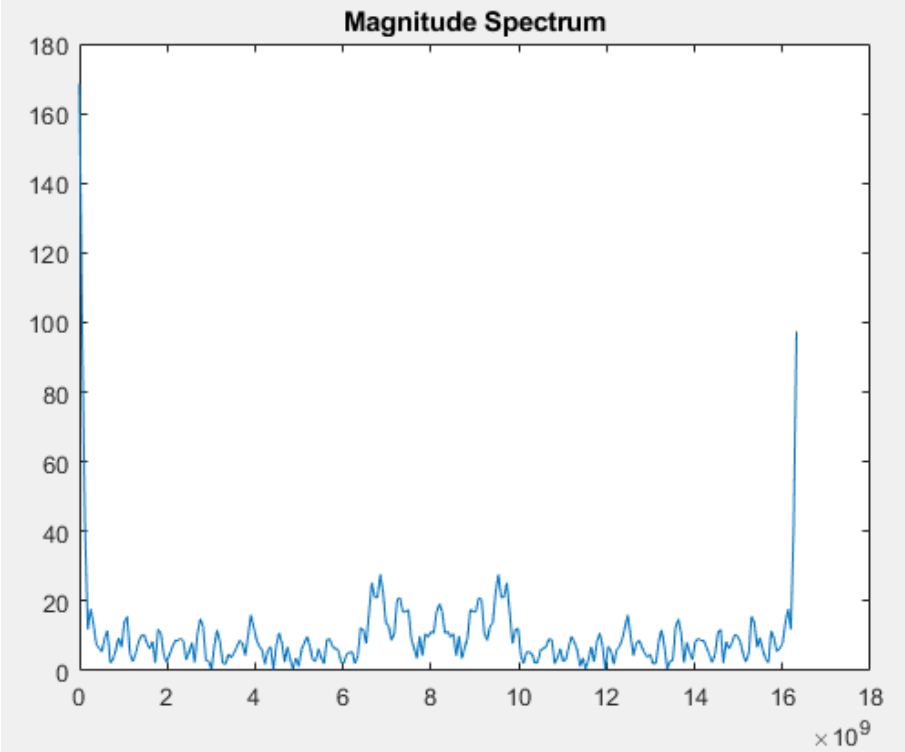


6 Difference Scan Plot



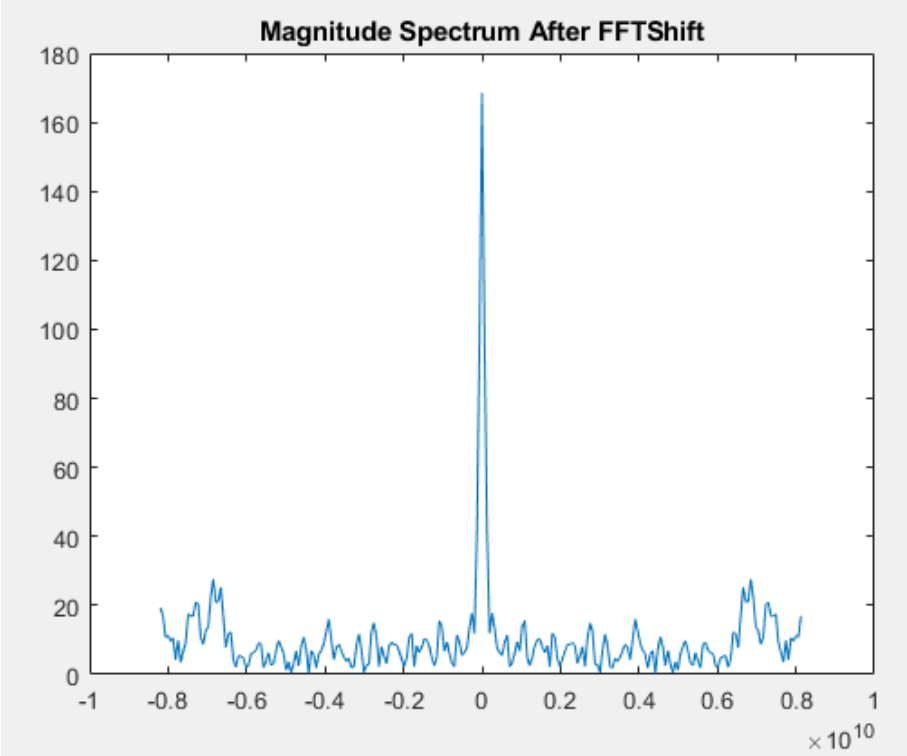
The peaks between 0 and 0.4 meters are likely myself. The peak at 0.6 is likely the classmate beside me. The peak at about 0.8 meters is the target.

6 Magnitude Spectrum Plot



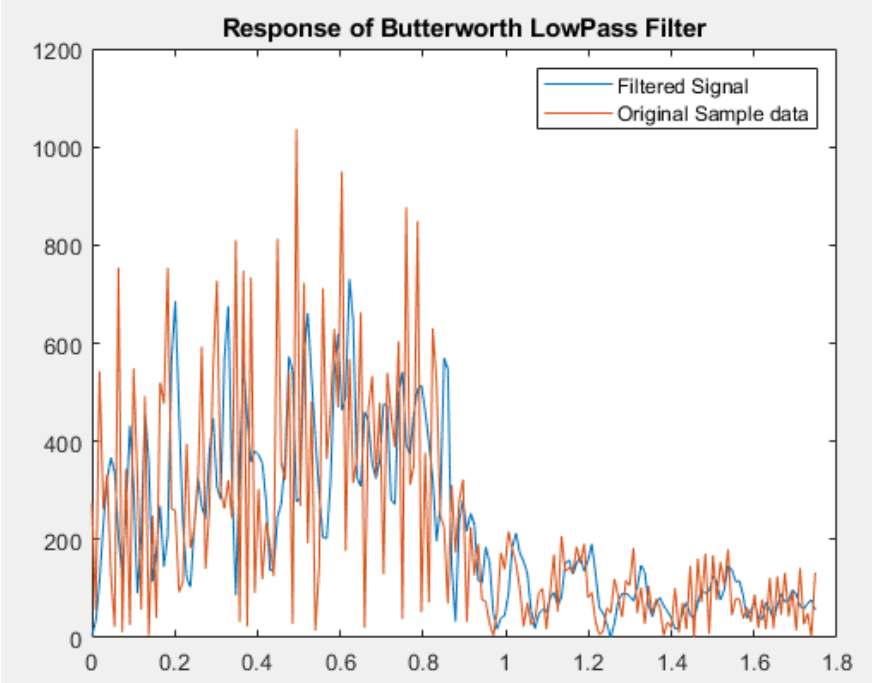
Frequencies Present: 0 to 16.3 GHz

6 Shifted Magnitude Spectrum Plot



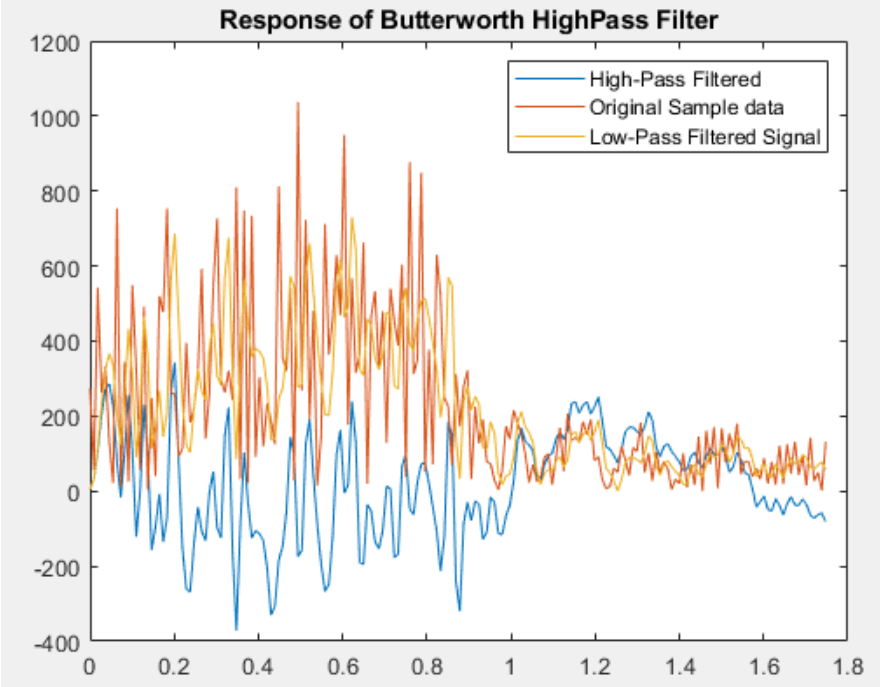
Frequencies Present: -8.15 GHz to 8.15 GHz

7 Low Pass Filtered Plot (Butterworth)



The filtered signal in the time domain has less noise compared to the original signal. The number of peaks is also reduced, as well as the height of the larger peaks.

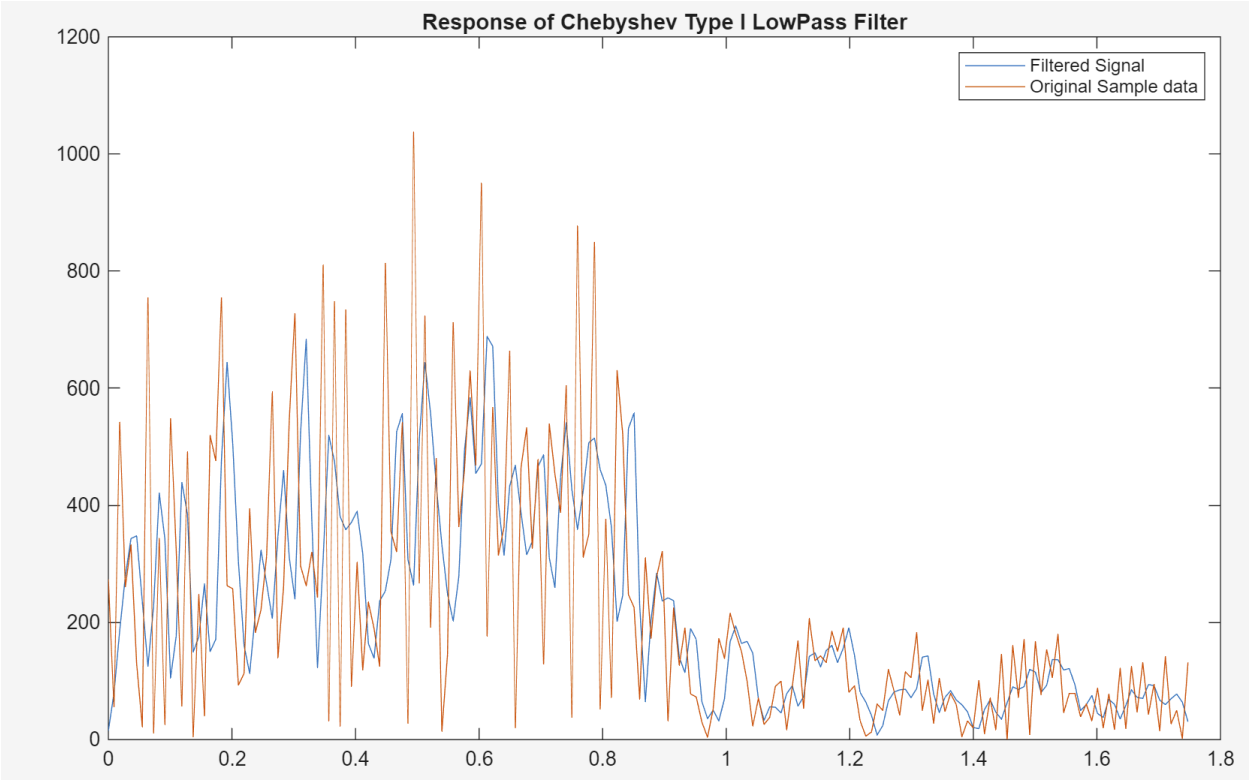
8 High Pass Filtered Plot (Butterworth)



The high pass filtered signal in the time domain has less noise compared to the original signal. It also has a lower gain on average, with the highest gain occurring at the target value at around 1

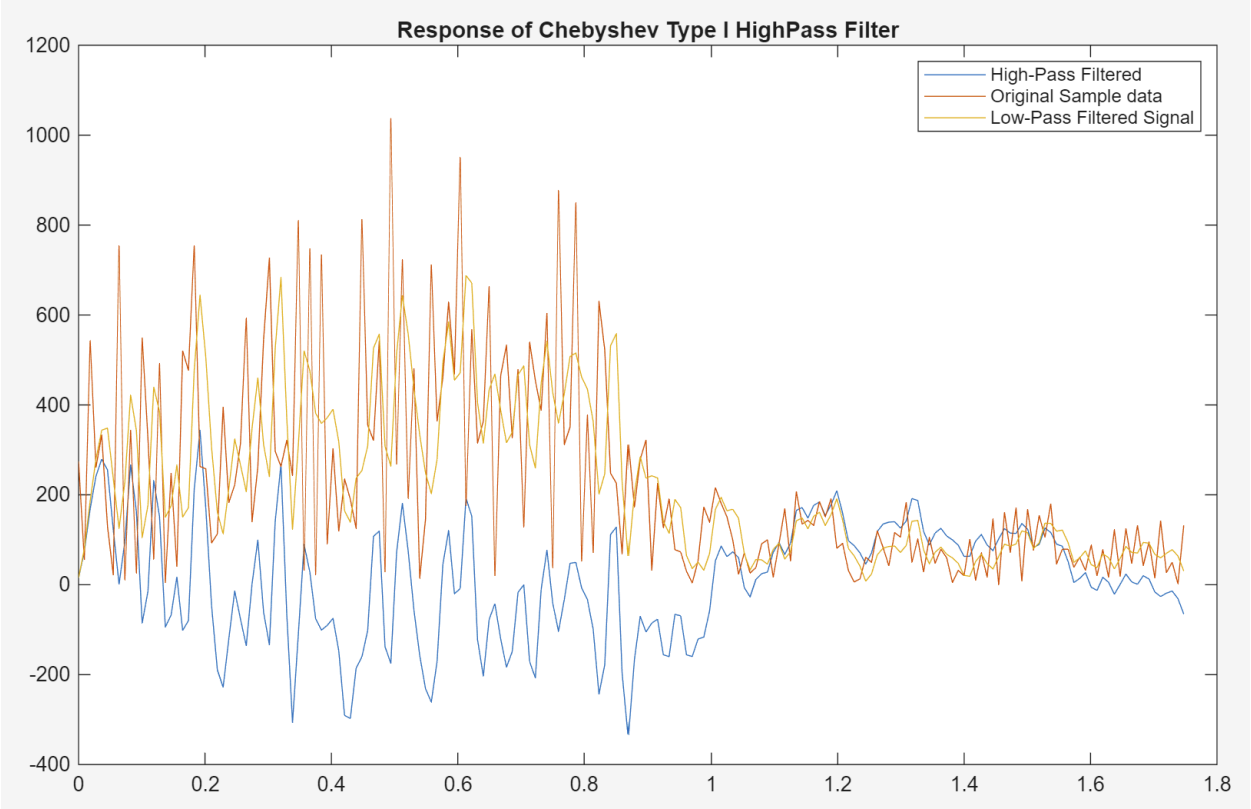
meter. The values at and beyond 1 meter are much higher than those before 1 meter on average.

9 Low Pass Filtered Plot (Chebyshev I)



The low pass filtered signal in the time domain has less noise compared to the original signal. It also has a lower value on average compared to the original.

9 High Pass Filtered Plot (Chebyshev I)



Like the previous filters, the high pass filtered signal in the time domain has less noise compared to the original signal. It also has a lower gain on average, with the highest gain occurring at the target value at around 1 meter. The values at and beyond 1 meter are much higher than those before 1 meter on average.

Questions:

1. The apparent frequency content near and beyond ±8.2 GHz is inaccurate because the Nyquist folding frequency limits valid spectral data strictly to $\pm\frac{f_s}{2}$ (around $\pm$8.15 GHz). Any components appearing beyond this boundary are mathematical artifacts caused by aliasing and improper frequency-axis indexing in the FFT calculation, where periodic boundary bins wrap around and distort the edges of the baseband spectrum.
2. The fftshift command rearranges the FFT output by moving the zero-frequency (DC) component from the beginning of the vector to the center. Plot 6.c.i displays an unshifted spectrum with 0 Hz at the far-left origin and frequencies spanning from 0 to 16.3 GHz, placing negative frequency components at the far-right edge. In contrast, plot 6.c.iii uses fftshift to center 0 Hz in the middle, displaying a symmetric two-sided spectrum ranging from -8.15 GHz to +8.15 GHz.
3. In the frequency domain, the low-pass filter directly blocks fast, high-frequency components above a chosen cutoff point while letting lower frequencies pass through unchanged. On a magnitude spectrum plot, this causes the higher frequency regions to drop down near zero, leaving only a clean band of low-frequency energy. In the time

domain, this action smooths out the signal by flattening sharp spikes, reducing noisy bumps, and introducing a tiny time delay that shifts the filtered peaks slightly after the originals.

4. In the frequency domain, the high-pass filter blocks low-frequency components below a chosen cutoff point, including the main DC offset, while allowing higher frequencies and rapid changes to pass through untouched. On a magnitude spectrum plot, the center area around 0 Hz drops near zero, leaving energy concentrated only in the outer high-frequency regions. In the time domain, this removes the signal's baseline trend and slow overall shifts, which causes the filtered signal to oscillate symmetrically around zero with both positive and negative values while preserving only the rapid, sharp fluctuations and detail.
5. To achieve the same cutoff sharpness, a Chebyshev Type I filter requires a lower filter order than a Butterworth filter. The main difference in performance is that the Butterworth filter gives a completely smooth response without distorting amplitudes, while the Chebyshev Type I filter has a sharper cutoff but introduces small ripples and fluctuations in the passband. As a result, the Chebyshev filter cuts off unwanted frequencies faster, but the Butterworth filter maintains a cleaner, smoother overall signal shape.